



“partial” rescaling, trade-off

Unbalanced OT: Computation

- Adding entropy regularization:

$$\min_{\pi \geq 0} \langle C, \pi \rangle + \tau_1 D_\psi(\pi \mathbf{1}_m \mid \mathbf{a}) + \tau_2 D_\psi(P^\top \mathbf{1}_n \mid \mathbf{b}) - \varepsilon H(\pi)$$

- UBOT can be solved via **modified Sinkhorn iterations**:

$$\mathbf{u} \leftarrow (\mathbf{a} \oslash \mathbf{K}\mathbf{v})^{\frac{\tau_1}{\tau_1 + \varepsilon}}, \quad \mathbf{v} \leftarrow (\mathbf{b} \oslash \mathbf{K}^\top \mathbf{u})^{\frac{\tau_2}{\tau_2 + \varepsilon}}$$

- Exercise: show that $\pi = \text{diag}(\mathbf{u})\mathbf{K}\text{diag}(\mathbf{v})$ might not have marginals $\mathbf{a}, \mathbf{b}$

Proof: Let $\xi_1 = \frac{\tau_1}{\tau_1 + \varepsilon}$. We have: $\pi \mathbf{1} = \mathbf{u} \odot (\mathbf{K}\mathbf{v}) = \left(\frac{\mathbf{a}}{\mathbf{K}\mathbf{v}}\right)^{\xi_1} \odot (\mathbf{K}\mathbf{v}) = \mathbf{a}^{\xi_1} \odot (\mathbf{K}\mathbf{v})^{1-\xi_1}$. Idem for $\mathbf{b}$.

- Intuition:**

- Exponents $\xi_1, \xi_2 \in (0, 1)$ yield **“partial” rescaling, trade-off** between marginals and current row/col sums
- Smaller $\tau_1 \rightarrow$ smaller $\xi_1 \rightarrow (\mathbf{K}\mathbf{v})^{1-\xi_1}$ dominates $\mathbf{a}^{\xi_1} \rightarrow \pi$'s marginal further away from $\mathbf{a}$
- Converges to regular Sinkhorn iterations as $\tau_i \rightarrow \infty$ (i.e. $\xi_i \rightarrow 1$).

Gromov-Wasserstein

